

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN Stack

M1:Practical – II

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Date of Assignment: 30/06/2026	Date/Time of Submission: 30/06/2026

2. Core JavaScript Concepts for MERN

a. Create functions and call them with parameters.

Code:

```
function calculateSalary(name, basicSalary){
    let bonus = 5000;
    let totalSalary = basicSalary + bonus;

    console.log("Employee Name:", name);
    console.log("Basic Salary:", basicSalary);
    console.log("Bonus:", bonus);
    console.log("Total Salary:", totalSalary);
}
calculateSalary("Anu", 50000);
console.log("~Asmitha");
```

Output:

```
PS C:\Users\itlab\Desktop\Asmitha\MERN STACK> node prac-2a
Employee Name: Anu
Basic Salary: 50000
Bonus: 5000
Total Salary: 55000
~Asmitha
```

Code:

```
function generateBill(product, quantity, price)
{
    let total = quantity * price;
    let bill = total + (total * 0.09);

    console.log("Product:", product);
    console.log("Quantity:", quantity);
    console.log("Price per item:", price);
    console.log("Total Bill", bill);
}
generateBill("Laptop", 2, 50000)
console.log("~Asmitha");
```

Output:

```
PS C:\Users\itlab\Desktop\Asmitha\MERN STACK> node prac-2a
Product: Laptop
Quantity: 2
Price per item: 50000
Total Bill 109000
~Asmitha
```

b. Work with arrays and objects in JavaScript.

Code:

```
let fruits = ["Apple", "Banana", "Cherry", "Durian"];
let student = {
  name: "Asmitha",
  age: 21,
  city: "Mumbai"
};
console.log("Fruits:", fruits);
console.log("Second Fruit:", fruits[2]);

console.log("Student Name:", student.name);
console.log("Student Age:", student.age);
console.log("Student City:", student.city);
console.log("~Asmitha")
```

Output:

```
PS C:\Users\itlab\Desktop\Asmitha\MERN STACK> node prac-2b
Fruits: [ 'Apple', 'Banana', 'Cherry', 'Durian' ]
Second Fruit: Cherry
Student Name: Asmitha
Student Age: 21
Student City: Mumbai
~Asmitha
```

c. Demonstrate use of arrow functions.

Code:

```
let add = (a, b) => {
  return a + b;
};
let sub = (a, b) => {
  return a - b;
};
let mult = (a, b) => {
  return a * b;
};

console.log("Addition =", add(10,20));
console.log("Subtraction =", sub(10,20));
console.log("Multiplication =", mult(10,20));
console.log("~Asmitha")
```

Output:

```

● PS C:\Users\itlab\Desktop\Asmitha\MERN STACK> node prac-2c
Addition = 30
Subtraction = -10
Multiplication = 200
~Asmitha

```

d. Write a program using array methods (map, filter, reduce).

Code:

```

let numbers = [10, 20, 30, 40, 50];
//map
let doubled = numbers.map(num => num * 2);
//filter
let greater25 = numbers.filter(num => num > 25);
//reduce
let total = numbers.reduce((sum, num) => sum + num, 0);
console.log("Original Array:", numbers);
console.log("Map:", doubled);
console.log("Filter:", greater25);
console.log("Reduce:", total);
console.log("~Asmitha")

```

Output:

```

● PS C:\Users\itlab\Desktop\Asmitha\MERN STACK> node prac-2d
Original Array: [ 10, 20, 30, 40, 50 ]
Map: [ 20, 40, 60, 80, 100 ]
Filter: [ 30, 40, 50 ]
Reduce: 150
~Asmitha

```

e. Create a program that manipulates objects and displays output.

Code:

```

let student = {
  name: "Asmitha",
  rollno: 2,
  marks: 95
};
console.log("Original Object:");
console.log(student);
//accessing property
console.log("\nAccessing property:");
console.log("Name:", student.name);
console.log("Marks:", student.marks);
//updating property
student.marks = 95;
console.log("\nAfter Updating Marks: \n", student);
//adding new property
student.city = "Mumbai";
console.log("\nAfter Adding: \n", student);
//delete property
delete student.rollno;
console.log("\nAfter deleteing rollno: \n", student);
console.log("~Asmitha")

```

Output:

```
PS C:\Users\itlab\Desktop\Asmitha\MERN STACK> node prac-2e
Original Object:
{ name: 'Asmitha', rollno: 2, marks: 95 }

Accessing property:
Name: Asmitha
Marks: 95

After Updating Marks:
{ name: 'Asmitha', rollno: 2, marks: 95 }

After Adding:
{ name: 'Asmitha', rollno: 2, marks: 95, city: 'Mumbai' }

After deleteing rollno:
{ name: 'Asmitha', marks: 95, city: 'Mumbai' }
~Asmitha
```

f. Implement a small JavaScript program combining functions, arrays, and objects.

Code:

```
let students = [
  {
    rollno: 101,
    name: "Asmitha",
    marks: 95
  },
  {
    rollno: 102,
    name: "Anu",
    marks: 100
  },
  {
    rollno: 103,
    name: "Anju",
    marks: 90
  }
];
```

```
//function to display student details
function display(s)
{
  console.log("Student Details:");
  console.log("-----")
  for(let stud of s)
  {
    console.log(`Roll No    : ${stud.rollno}`);
    console.log(`Name      : ${stud.name}`);
    console.log(`Marks     : ${stud.marks}`);
    console.log(`-----`);
  }
}
//function call
display(students);
console.log("~Asmitha")
```

Output:

```
● PS C:\Users\itlab\Desktop\Asmitha\MERN STACK> node prac-2f
Student Details:
-----
Roll No    : 101
Name       : Asmitha
Marks      : 95
-----
Roll No    : 102
Name       : Anu
Marks      : 100
-----
Roll No    : 103
Name       : Anju
Marks      : 90
-----
~Asmitha
```